

Joshua Min

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EDUCATION

University of California, Berkeley

Bachelor of Arts in Data Science

Berkeley, CA

Aug. 2025 – May 2029

EXPERIENCE

Autonomous Vehicle Developer

UC Berkeley Robot Open Autonomous Racing

July 2024 – Aug. 2024

Berkeley, CA

- Engineered autonomous racing agents using Python, TensorFlow, and OpenCV, focusing on control systems
- Trained agents over 300+ simulation episodes on Monza racetrack to optimize reward-based performance
- Implemented Deep Q-Learning, PID controllers, and multilayer perceptrons to control high-speed track environments
- Collaborated with a team of student engineers on racing strategies and model generalization
- Achieved top 10 placement in ROAR Competition by minimizing track deviation and improving throttle control

Front-End Development & Product Intern

Mapt.co

June 2023 – Aug. 2023

Asheville, NC

- Redesigned and enhanced app and website UI/UX to improve navigation and responsiveness across desktop and mobile platforms
- Implemented interactive front-end features using HTML, CSS, and JavaScript frameworks to improve user engagement
- Collaborated with CPO and CTO to identify and resolve bugs, streamline code, and test new releases
- Contributed to product roadmap and feature prioritization through cross-functional coordination and user feedback

Hackathon Project Lead

Dublin Hack Club

Aug. 2022 – May 2025

Dublin, CA

- Led school competition team at Innovate Hacks to build an interactive arithmetic dice game for children using Python and PyGame
- Developed modular game logic with scoring, animations, and adaptive difficulty
- Managed version control and team workflows using Git and GitHub, including branching strategies and code reviews to maintain team's code quality
- Collaborated in coding workshops to build projects and explore Python libraries

PROJECTS

MusicGen | *Python, TensorFlow, Keras, music21*

Jan. 2024 – Feb. 2025

- Built and trained an LSTM model to generate melodies in pop, classical, and video game styles
- Used music21 and TensorFlow for sequence modeling and multi-instrument composition
- Processed 600 MIDI files into a clean, labeled dataset for training and evaluation
- Generated 70+ unique pieces, including a case study on 48 Chopin works for stylistic analysis

NeuroMIND | *Python, OpenCV, Tensorflow, Matplotlib, Jupyter*

May 2024 – April 2025

- Developed a CNN model to classify Alzheimer's stages using 6,000+ brain MRI scans with 92% accuracy
- Applied data augmentation and cross-validation to optimize model performance and generalizability
- Visualized key features with Grad-CAM heatmaps to interpret spatial model attention
- Engineered preprocessing pipeline for normalization and dimensionality reduction of MRI data

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, HTML/CSS

Developer Tools: Git, GitHub, VS Code, Visual Studio, PyCharm, Jupyter

Libraries: pandas, NumPy, Matplotlib, PyTorch, Tensorflow, seaborn, OpenCV, scikit-learn